

Ioana Oriana Bercea

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Positions

- 1/10/2023– **Assistant Professor (tenure-track)**,
present DIVISION OF THEORETICAL COMPUTER SCIENCE
Department of Computer Science, School of Electrical Engineering and Computer Science
KTH ROYAL INSTITUTE OF TECHNOLOGY, Stockholm, Sweden
- 1/05/2021– **Postdoc**,
30/09/2023 BASIC ALGORITHMS RESEARCH COPENHAGEN (BARC)
Department of Computer Science
IT UNIVERSITY OF COPENHAGEN, Copenhagen, Denmark
Host: Thore Husfeldt
- 1/09/2018– **Postdoc**,
30/04/2021 School of Electrical Engineering
TEL AVIV UNIVERSITY, Tel Aviv, Israel
Host: Guy Even

Education

- 03/09/2013– **Ph.D. in Computer Science**,
17/08/2018 UNIVERSITY OF MARYLAND, College Park, MD, USA
Thesis: *Approximation Algorithms for Geometric Clustering and Touring Problems*
Advisor: Samir Khuller
- 30/08/2010– **M.Sc. in Computer Science**,
19/05/2013 UNIVERSITY OF MARYLAND, College Park, MD, USA
Advisor: Aravind Srinivasan
- 25/09/2006– **B.Sc. in Mathematics (Honors) and B.Sc. in Computer Science**,
12/06/2010 UNIVERSITY OF CHICAGO, Chicago, IL, USA
Four-year scholarship for tuition, housing and stipend (unconditioned on GPA)

Honors and Awards

- 2026 **Best Research Paper Honorable Mention**, SIGMOD
- 2026 **Research Highlight**, SIGMOD
- 2025 **Best Research Paper**, VLDB
- 2023 **Best Artifact**, ACM SIGMOD
- 2023 **Future Digileader**, DIGITAL FUTURES
- 2014–2015 **Outstanding Graduate Assistant Award**, UNIVERSITY OF MARYLAND
Top 2% of all UMD Graduate Assistants
- 2010–2012 **Dean's Fellowship**, UNIVERSITY OF MARYLAND
- 2006–2008, **Dean's List**, UNIVERSITY OF CHICAGO
2009–2010

Grants

- 2025 – 2028 **Starting Grant**, SWEDISH RESEARCH COUNCIL
Natural and engineering sciences, 1 100 000 SEK/year

Refereed Conference Articles (Author order alphabetical, unless stated otherwise)

- ICML'26 **Instance-Specific Approximation Ratios for Correlation Clustering and Max-Cut**
Sebastian Lüderssen, Ioana-Oriana Bercea, Stefan Neumannn (*authors in order of contribution*)
In: *International Conference on Machine Learning*, (ICML)
- SIGMOD'26 **Sublime: Sublinear Error & Space for Unbounded Skewed Streams**
Navid Eslami, Ioana O. Bercea, Rasmus Pagh, Niv Dayan (*authors in order of contribution*)
In: *International Conference on Management of Data*, (SIGMOD)
Best Research Paper Honorable Mention
- VLDB'25 **Diva: Dynamic Range Filter for Var-Length Keys and Queries**
Navid Eslami, Ioana O. Bercea, Niv Dayan (*authors in order of contribution*)
In: *Very Large Data Bases Conference*, (VLDB)
Best Research Paper
SIGMOD 2026 Research Highlight
- ICALP'25 **Algorithms for the Diverse-k-SAT problem: the geometry of satisfying assignments**
Per Austrin, Ioana O. Bercea, Mayank Goswami, Nutan Limaye, and Adarsh Srinivasan
In: *EATCS International Colloquium on Automata, Languages, and Programming*, (ICALP)
Invited talk to Highlights of Algorithms (HALG) 2026
- CIAC'25 **Dynamic Filter and Retrieval with One Access to Modifiable Memory**
Ioana O. Bercea, Guy Even, Tomer Even and Gabriel Marques Domingues
In: *International Conference on Algorithms and Complexity*, (CIAC)
- ESA'24 **Online sorting and online TSP: randomized, stochastic, and high-dimensional**
Mikkel Abrahamsen, Ioana O. Bercea, Lorenzo Beretta, Jonas Klausen, László Kozma
In: *European Symposium on Algorithms*, (ESA)
- VLDB'24 **Aleph Filter: To Infinity in Constant Time**
Niv Dayan, Ioana O. Bercea, Rasmus Pagh (*authors in order of contribution*) (*authors in order of contribution*)
In: *Very Large Data Bases Conference*, (VLDB)
- SWAT'24 **Daisy Bloom Filters**
Ioana O. Bercea, Jakob Bæk Tejs Houen, Rasmus Pagh
In: *19th Scandinavian Symposium and Workshops on Algorithm Theory*, (SWAT)
- FOCS'23 **Locally Uniform Hashing**
Ioana O. Bercea, Lorenzo Beretta, Jonas Klausen, Jakob Bæk Tejs Houen, Mikkel Thorup
In: *64th IEEE Symposium on Foundations of Computer Science*, (FOCS)
- SIGMOD'23 **InfiniFilter: Expanding Filters to Infinity and Beyond**
Niv Dayan, Ioana O. Bercea, Pedro Reviriego, Rasmus Pagh (*authors in order of contribution*)
In: *International Conference on Management of Data*, (SIGMOD)
Best Artifact
- STOC'22 **An Extendable Data Structure for Incremental Stable Perfect Hashing**
Ioana O. Bercea, Guy Even
In: *54th Annual ACM Symposium on Theory of Computing*, (STOC)
- WADS'21 **Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *17th Algorithms and Data Structures Symposium*, (WADS)
Invited to the special issue of Algorithmica
- CIAC'21 **Upper Tail Analysis of Bucket Sort and Random Tries**
Ioana O. Bercea, Guy Even
In: *12th International Conference on Algorithms and Complexity*, (CIAC)
Invited to the special issue of Theoretical Computer Science

- SWAT'20 **A Dynamic Space-Efficient Filter with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *17th Scandinavian Symposium and Workshops on Algorithm Theory, (SWAT)*, pp. 11:1-11:17
- APPROX'19 **On the cost of essentially fair clusterings**
Ioana O. Bercea, Martin Groß, Samir Khuller, Aounon Kumar, Clemens Rösner, Daniel R. Schmidt, Melanie Schmidt
In: *22nd International Workshop on Approximation Algorithms for Combinatorial Optimization Problems, (APPROX)*, pp. 18:1-18:22
- CCCG'18 **Improved Bounds for the Traveling Salesman Problem with Neighborhoods on Uniform Disks**
Ioana O. Bercea
In: *30th Canadian Conference on Computational Geometry, (CCCG)*, pp.129-141
- CCCG'16 **Minimizing Uncertainty through Sensor Placement with Angle Constraints**
Ioana O. Bercea, Volkan Isler, Samir Khuller
In: *28th Canadian Conference on Computational Geometry, (CCCG)*, pp. 287-294
- SPAA'14 **On Computing Maximal Independent Sets of Hypergraphs in Parallel**
Ioana O. Bercea, Navin Goyal, David G. Harris, Aravind Srinivasan
In: *26th ACM Symposium on Parallelism in Algorithms and Architectures, (SPAA)*, pp. 42-50
Invited to the special issue of TOPC

Journal Articles (Author order alphabetical, unless stated otherwise)

- 2026 **Dynamic Range Filtering Beyond Worst-Case Bounds**
Navid Eslami, Ioana O. Bercea, Niv Dayan (*authors in order of contribution*)
In: *ACM SIGMOD Record*, Volume 55, Issue 1
- 2022 **Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *Algorithmica*, Special issue on WADS 2021
- 2021 **Upper Tail Analysis of Bucket Sort and Random Tries**
Ioana O. Bercea, Guy Even
In: *Theoretical Computer Science*, Special issue on CIAC 2021
- 2016 **On Computing Maximal Independent Sets of Hypergraphs in Parallel**
Ioana O. Bercea, Navin Goyal, David G. Harris, Aravind Srinivasan
In: *ACM Transactions on Parallel Computing*, Special issue on SPAA 2014

Invited Workshops

- Feb. 2026 NII SHONAN: *New Era of Data Structures: From Strings to Graphs, and more*
- Aug. 2025 DAGSTUHL : *Software Performance Engineering*
- May 2025 DAGSTUHL : *Adaptive and Scalable Data Structures*
- May 2025 DAGSTUHL : *Learned Predictions for Data Structures and Running Time*
- March 2025 NII SHONAN: *Algorithmic tools and their applications in emerging models of computation*
- July 2024 NU SINGAPORE: *Algorithms and Data Structures Today*
- Sept 2023 BERTINORO : *Algorithmic Aspects of Clustering and Related Problems*
- May 2023 DAGSTUHL : *Scalable Data Structures*
- March 2023 NII SHONAN : *New Frontiers in Locality in Computation*
- Feb. 2021 DAGSTUHL : *Scalable Data Structures*
- Oct. 2019 IGAFIT: *Workshop for Postdoctoral Researchers in Algorithms*
- Sept. 2019 GI-DAGSTUHL : *Algorithms for Big Data*
- March 2019 DAGSTUHL : *Theoretical Foundations of Storage Systems*

Jan. 2019 DAGSTUHL: *Data Structures for the Cloud and External Memory Data*
Oct. 2018 *7th French-Israeli Workshop on Foundations of Computer Science*

Invited Talks

Dec 2025 A CELEBRATION OF TCS, Sydney, Australia
Oct 2025 WAVE (WOMEN IN ALGORITHMS, VENTURE INTO EXPLORATION), Copenhagen, Denmark
June 2025 BOCCONI THEORY DAY, Milan, Italy
Dec 2024 AUSTRALASIAN SUMMER SCHOOL ON RECENT TRENDS IN ALGORITHMS, Sydney, Australia
July 2023 ROMANIAN ALGORITHMS DAYS, virtual

Supervision

2021-2025 **Jonas Østergaard Klausen**
PhD, UNIVERSITY OF COPENHAGEN
co-supervised with Mikkel Thorup, Jacob Holm
→ postdoc at MPI Saarbrücken

2025 **Lisa Schmierer**
Master's thesis, TECHNICAL UNIVERSITY OF MUNICH (TUM)

2025 **Rani Eydfinnsson ð Bø**
Master's thesis, TECHNICAL UNIVERSITY OF DENMARK (DTU)
co-supervised with Eva Rotenberg

Summer 2024, 2025 **Max Harrison, Brandy Rydberg Paltser, Eira Larsson**
Digital Futures Internship, KTH

Spring 2023 **Jens Birk Andersen, Malthe Mathias Mølgaard Larsen**
BSc Thesis, IT UNIVERSITY OF COPENHAGEN
co-supervised with Martin Aumüller

Teaching

Fall 2023 – **Course Responsible, Teacher for DD2440 Advanced Algorithms, MSc,**
KTH ROYAL INSTITUTE OF TECHNOLOGY, Department of Computer Science

Summer 2022 **Co-Instructor for Algorithms and Data Structures BSc, MSc,**
& 2023 IT UNIVERSITY OF COPENHAGEN, Department of Computer Science

Spring 2016 **Co-Instructor for CMSC 122: Intro to Computer Programming via the Web,**
UNIVERSITY OF MARYLAND, Department of Computer Science

2010–2018 **Graduate Teaching Assistant,**
UNIVERSITY OF MARYLAND, Department of Computer Science
D = Leading discussion sections, presenting new material, reviewing, quizzes, grading, office hours; G = Grading and office hours

CMSC 250: Discrete Structures,
Fall 2011(D), Spring 2014(D), Spring 2015(D), Fall 2015(D), Fall 2016(D), Spring 2017(G)

CMSC 131/132: Object Oriented Programming I, II,
Fall 2010 (D), Fall 2013(D), Spring 2018(G)

CMSC 216: Introduction to Computer Systems,
Spring 2011(G), Summer 2011(D)

CMSC 351: Introduction to Algorithms,
Spring 2012(G)

CMSC 122: Intro to Computer Programming via the Web,
Fall 2017(G)

2009–2010 **Junior Tutor for Elementary Functions and Analysis 1,2,3,**
UNIVERSITY OF CHICAGO, Mathematics Department

Service

- 2024 – **Co-editor** ALGORITHMS COLUMN FOR THE BULLETIN OF EATCS (*with Thomas Erlebach*)
- 2025 – 26 **Local chair** HIGHLIGHTS OF ALGORITHMS (*HALG 2026*)
- 2024 – 25 **Guest journal editor** THEORY OF COMPUTING (*Special issue on RANDOM 2024*)
- 2025 **Program Committee co-chair** SOSA 2025 (*with Rasmus Pagh*)
- Program Committee member** ESA 2026 (TRACK S), SEA 2026, COCOON 2026, SWAT 2026, ICALP 2026, STACS 2026, SODA 2026, ESA 2025 (TRACK A), RANDOM 2024, ISAAC 2024, SOSA 2024, ICDCN 2020
- Conference Reviewer**
- 2025 FOCS, IPCO
- 2024 FOCS, STOC, SODA, ICALP, ITCS, SWAT, SPAA, ESA, MFCS, SOFSEM
- 2023 FOCS, SODA, ICALP, ISAAC, FCT, SOSA, ALENEX
- 2022 FOCS, STOC, APPROX
- 2020 – 21 WAOA'21, ESA'21, SPAA'21, WADS'21, SODA'20, STACS'20
- < 2020 ICALP '24, SWAT'24, SPAA'24, STOC'24, ITCS'24, SODA'24
- Journal Reviewer** *SIAM Journal on Computing, TheoretCS, Journal of Computer Science and Technology, Transactions on Computers, European Journal of Operational Research, Theoretical Computer Science, Networks, ACM Transactions on Sensor Networks, The Visual Computer*
- 2015–2016 **Graduate Student Representative**,
UNIVERSITY OF MARYLAND, Computer Science Department Council
- 2012–2014 **Graduate Student Representative**,
UNIVERSITY OF MARYLAND, Computer Science Department Education Committee

Talks

Locally Uniform Hashing

- Jan. 2026 UNIVERSITY OF TORONTO: *Theory seminar*
- Sept. 2025 UPPSALA UNIVERSITY: *Probabilities and Statistics seminar*
- Aug. 2025 TU WIEN
- June 2025 BOCCONI THEORY DAY
- May 2022 ALGORITHMIC RESEARCH COOPERATION AROUND ØRESUND (ARCO)
- March 2025 NII TOKYO
- March 2025 TOKYO TECH
- July 2025 NU SINGAPORE: *Algorithms and Data Structures Today*
- July 2023 ROMANIAN ALGORITHMS DAYS

Locality in Data Structures

- May 2023 AARHUS UNIVERSITY, *Algorithms and Complexity Theory Seminar*
- March 2023 NII SHONAN MEETING: *New Frontiers in Locality in Computation*

Daisy Bloom Filters

- June 2024 SWAT
- Nov. 2022 ALGORITHMIC RESEARCH COOPERATION AROUND ØRESUND (ARCO)

An Extendable Data Structure for Incremental Stable Perfect Hashing

- Oct. 2025 WAVE (WOMEN IN ALGORITHMS, VENTURE INTO EXPLORATION)
- May 2023 DAGSTUHL SEMINAR: *Scalable Data Structures*
- July 2022 STOC
- June 2022 HIGHLIGHTS OF ALGORITHMS (HALG) (short)

Dictionaries et al.

- Sept. 2021 DIGITAL RESEARCH CENTER DENMARK (DIREC)
May 2021 BARC, *Seminar*
March 2021 BAR-ILAN UNIVERSITY, *Computer Science Department Colloquium*
Feb. 2021 DAGSTUHL SEMINAR: *Scalable Data Structures*

Upper Tail Analysis of Bucket Sort and Random Tries

- June 2020 20TH HAIFA WORKSHOP ON GRAPH THEORY, COMBINATORICS AND ALGORITHMS

Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations

- Aug. 2021 WADS

A Dynamic Space-Efficient Filter with Constant Time Operations

- June 2020 SWAT
Dec. 2019 UTRECHT UNIVERSITY, Algorithms Seminar
Sept. 2019 GI-DAGSTUHL SEMINAR: Algorithms for Big Data
July 2019 WORKSHOP ON LOCAL ALGORITHMS (WOLA)

The Descent of Cuckoos, and Selection in Relation to Nests

- June 2019 HIGHLIGHTS OF ALGORITHMS (HALG) (short)
June 2019 19TH HAIFA WORKSHOP ON GRAPH THEORY, COMBINATORICS AND ALGORITHMS
May 2019 ISRAELI NETWORKING DAY
March 2019 MAX PLANCK INSTITUTE FOR INFORMATICS, Theory Seminar
Jan. 2019 DAGSTUHL SEMINAR: *Data Structures for the Cloud and External Memory Data*

Improved Bounds for the Traveling Salesman Problem with Neighborhoods on Uniform Disks

- Nov. 2018 TEL AVIV UNIVERSITY, Computational Geometry Seminar
Oct. 2018 *7th French-Israeli Workshop on Foundations of Computer Science*
Aug. 2018 CCCG
April 2018 CENTRA DE RECERCA MATEMATICA, IRP Program

Minimizing Uncertainty through Sensor Placement with Angle Constraints

- Aug. 2016 CCCG

On Computing Maximal Independent Sets of Hypergraphs in Parallel

- June 2014 SPAA